

Dhruv Agarwal

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SKILLS

Languages:	C++, JavaScript, TypeScript, Python, SQL
Backend:	Node.js, Express.js, Django, Django REST Framework, REST APIs, API Gateways, WebSockets, Socket.IO, Redis, LiveKit
Frontend and AI:	React, Next.js, Redux Toolkit, Tailwind CSS, Gemini API, Groq API, LangChain.js, LangGraph.js, Tavily API, LLM APIs
Databases:	MongoDB, Mongoose, MySQL, PostgreSQL, MongoDB Transactions, Database Indexing
Security and Testing:	JWT, RBAC, API Key Authentication, Node.js Test Runner, Django TestCase, Postman
Tools and DevOps:	Git, GitHub, Postman, Docker, Docker Compose, Render, Vercel

PROJECTS

- Equilibrium | Multi-Agent AI Investment Research Platform** | [GitHub](#) | [Live](#)
 - Orchestrated a five-node LangGraph.js workflow (Research, Bull, Bear, Risk, Judge) with three parallel analysis agents, combining their outputs with a user-selected risk mandate to generate an Invest / Pass / Watch verdict.
 - Integrated Tavily news search and Alpha Vantage fundamentals in the research node to compile a factual, opinion-free dossier per company from live data.
 - Validated structured LLM outputs with Zod and saved research sessions in MongoDB for retrieval in a React dashboard.
Tech: Node.js, Express.js, LangGraph.js, Groq API, Tavily API, Alpha Vantage API, Zod, MongoDB, React
- LLM Access, Budget & Resilience Gateway** | [GitHub](#) | [Live](#)
 - Built 18 versioned REST endpoints and a React/Redux dashboard for account access, API-key management, wallet balances, and paginated usage history.
 - Implemented atomic credit reservation, settlement, and release using MongoDB transactions and integer micro-paise accounting across shared wallets and per-key budgets.
 - Validated authentication, billing logic, and privacy controls with 69 passing Node.js tests, including 429/5xx retries and timeout handling.
 - Kept Gemini credentials server-side; hashed passwords with scrypt and project keys with SHA-256, and capped provider calls at two attempts per request.
Tech: Node.js, Express.js, MongoDB, React, Redux Toolkit, Gemini API, JWT, RBAC, Docker
- VidConference | LiveKit-Powered Video Conferencing App** | [GitHub](#) | [Live](#)
 - Built meeting creation, scheduling, and join flows across Next.js and Django REST Framework; integrated LiveKit for audio/video, chat, and screen sharing.
 - Enforced host-only moderation with backend access-token checks; passed five Django API tests covering mute-all, participant ejection, non-host rejection, and audio-state persistence.
 - Implemented camera/microphone switching and meeting-link normalization; configured Docker Compose for LiveKit and a Render Blueprint for Django, PostgreSQL, and Next.js.
Tech: Django REST Framework, Next.js, React, LiveKit, PostgreSQL, Docker, Render
- Syntax Rush | Real-Time Multiplayer Coding Platform** | [GitHub](#) | [Live](#)
 - Built 10+ REST endpoints with JWT authentication and isolated coding sessions; checked request validation, access controls, and error responses using Postman.
 - Connected live coding sessions through WebSockets and Redis pub/sub, persisted data in MongoDB, and deployed the frontend and backend separately on Render.
Tech: Next.js, Tailwind CSS, Node.js, Express.js, MongoDB, WebSockets, Redis, JWT

CERTIFICATIONS

Privacy and Security in Online Social Media – NPTEL	Nov '25
Bring AI to Work Workshop – Google	Jun '25
MongoDB Advanced Schema Design Patterns and Antipatterns – MongoDB	Jun '25
Computer Communications Specialization – University of Colorado System	Nov '24

ACHIEVEMENTS

- Solved 600+ DSA problems across LeetCode, Codeforces, and GeeksforGeeks, including a 490-day daily LeetCode streak, demonstrating strong algorithmic thinking and system-optimization skills.
- Completed *Communication in the 21st Century Workplace* – University of California, Irvine Feb '26
- Participated in major hackathons: Build-a-Thon (Apna College, LPU) and Code-A-Haunt (Coding Blocks)

EDUCATION

Lovely Professional University Bachelor of Technology – Computer Science and Engineering; CGPA: 7.79	Punjab, India Aug '23–Present
Indra Prastha Academy Intermediate; Percentage: 76.6%	Babina, Jhansi Apr '21–Mar '23
St. Xavier's School Matriculation; Percentage: 84%	Babina, Jhansi Apr '19–Mar '21